FIRE RETARDANT COATINGS FOR STRUCTURES

MES-ST-SPC-0010

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Table of Contents

1. Introduction 3

1.1 Scope 3

1.2 Purpose 3

1.3 Conflicts and Deviations 4

2. Key Terms and Definitions 4

3. Referenced Documents 5

3.1 Statutory Regulations 5

3.2 Industry Codes and Standards 6

3.2.1 Local Codes and Standards (for Saudi Arabia) 6

3.2.2 International Codes and Standards 6

3.2.3 Alternative Design Codes 7

3.2.4 Insurer's Requirement 7

3.3 Ma'aden Standards 7

4. General Design Requirements 8

4.1 General 8

4.2 Supplier Drawing and Documentation 8

4.3 Safety, Reliability, Availability and Maintainability 8

4.4 Site Data 9

4.5 Design Life 9

4.6 Provision for Future Expansion 9

4.7 Standardization 9

4.8 Quality Assurance 9

5. Requirement Of FIRE-RETARDANT Coatings for Structures 10

5.1 Legislative Requirements 10

5.2 Fire Risk Assessment 10

5.3 System Design 11

6. Fire Retardant Coatings for Steel Structures 12

7. Fire Retardant-Treated Wood 14

8. Fire Retardant Coatings for Building Structures 15

1. INTRODUCTION

1.1 Scope

This document specifies the minimum requirements for Fire Retardant Coatings for Structures for all the Employer's projects.

The objective of the Fire-Retardant Coatings for Structures is to provide protection on the identified structures to resist fire damage long enough to safely shut down, evacuate the affected area and commence firefighting activities. The fire-retardant coating is intended to delay ignition and reduce the surface-burning rate of cellulosic fiber, cellular plastic and wood building material and to insulate steel structures to limit failure due to a reduction in mechanical properties.

This specification excludes:

- Fire barrier and requirements for fire rated walls.
- The requirements for sealing of electrical and mechanical services that penetrate through the fire rated walls or floors of switch rooms, cable rooms, control rooms and other buildings or structures.
- Blast-resistance requirements for buildings and structures.
- The fire rated coating of electrical cables.

The requirements for fire detection, alarm and protection systems are specifically excluded from this document.

Fire detection and alarm systems that detect and give warning of smoke and fires are included in the Fire Detection and Alarm Systems (General) Specification – MD-101-SMEM-EG-FI-SPC-0002.

Active fire protection systems are included in the Fire Protection Systems Specification – MD-101-SMEM-EG-FG-SPC-0003.

This document is intended to be used in conjunction with the referenced documents listed herein.

1.2 Purpose

The intention of this specification is to set the minimum engineering standard for Fire Retardant Coatings for Structures.

1.3 Conflicts and Deviations

Any deviation from the requirements stipulated in this document shall go through a waiver approval process as per Ma'aden Capital Investment System (MCIS) guidelines and shall be addressed to Ma'aden's Project Development & Engineering (PDE) team for their review and approval.

2. KEY TERMS AND DEFINITIONS

Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates
Contractor	Party responsible for supplying materials, plant, labor and supervisory services
Manufacturer	Party who produces and warrants performance of products, materials, and/or items provided in accordance with contract documents
Supplier	Party responsible for supplying concrete materials in accordance with contract documents
Engineer	The engineering design company that is appointed by Ma'aden for the project.
Fire Retardant Coating	A coating that reduces the flame spread of the tested combustible surfaces to which it is applied, by at least 50% or to a flame spread index value of 75 or less, whichever is the lesser value. The coated surface shall have a smoke developed index of not exceeding 200.
Fire Retardant-Treated Wood	A wood product impregnated with chemical during manufacture, and has a listed flame spread of 25 or less.
Fire Resistance Rating	The period of time a building element, component or assembly maintains the ability to confine a fire, continues to perform a given structural function, or both, as determined by the tests, or the methods based on tests described in section 4B.3 of SBC 801.

Flame Spread Index	The numerical value assigned to a material tested for the propagation of flame over a surface in accordance with ASTM E 84.
Intumescent Coating	A coating in which a chemical reaction occurs in passive materials, when exposed to high heat or direct flame impingement, that protects by expanding into an insulating layer of carbonaceous char material.
Site	Geographical location where services are to be performed.
Smoke-developed Index	The numerical value assigned to a material tested indicating a comparative measure derived from smoke obscuration data collected during the test for surface burning characteristics in accordance with ASTM E 84.
Supplier	Has the same meaning to Vendor, Contractor, Sub-Supplier or Sub-Contractor who are appointed to supply engineering equipment or services to the Employer.
SFRM	Sprayed Fire Resistant Materials
UL	Underwriters Laboratories

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest revision of the relevant standards and codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable. In general, the Contractor shall be responsible for full compliance with applicable Ma'aden Health, Safety and Environmental Standards (MSHEMs). Any deviation from the MSHEMs shall require written approval from the Employer.

3.1 Statutory Regulations

All work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia or local requirements of project Site. The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or

authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

All equipment and systems shall conform to applicable parts of the codes and standards, including but not limited to the following.

3.2.1 Local Codes and Standards (for Saudi Arabia)

All Ma'aden facilities shall comply with latest version of all Saudi Arabia applicable local codes and standards including but not limited to:

Particular attention is required to the requirements of the following:

SBC 801	Saudi Building Code – Fire Protection Requirements
SAF-4	HCIS Safety and Fire Protection Directives for Industrial Facilities – Fire Protection Systems
SAF-3	HCIS Safety and Fire Protection Directives for Industrial Facilities – Plant Buildings
SAF-9	HCIS Safety and Fire Protection Directives for Industrial Facilities – Fireproofing in Plants

For project locations other than Saudi Arabia, the Contractor shall follow the respective applicable local regulations, codes and standards

3.2.2 International Codes and Standards

NFPA 1	Fire Code
NFPA 101	Life Safety Code
ASTM E84 REV D	Standard Test Method for Surface Burning Characteristics of Building Materials
NFPA 703	Standard for Fire-Retardant-Treated Wood and Fire-Retardant Coatings for Building Materials
FM 1-51	Fire-Retardant Coatings and Paints for Interior Finish Materials

FM 1-61	Fire-Retardant Treated Wood
ISO 23932	Fire Safety Engineering – General Principles
API RP 752	Management of Hazards Associated with Location of Process Plant Permanent Buildings
API RP 2001	Fire Protection in Refineries
API PUBL 2218	Fireproofing Practices in Petroleum and Petrochemical Processing Plants
EN 2	Classification of Fires
ASTM E84	Standard Test Method for Surface Burning Characteristics of Building Materials
ISO 31000	Risk Management – Guidelines

3.2.3 Alternative Design Codes

Alternative design codes may be used for Fire Retardant Coatings for Structures design if approved by the Employer.

3.2.4 Insurer's Requirement

In addition to the requirements of the statutory authority, the Fire-Retardant Coatings for Structures design shall comply with the requirements of the Employer's Insurance Provider.

3.3 Ma'aden Standards

MD-101-SMEM-EG-FG-CRT-0001	Fire Protection Systems Design Criteria
MD-101-SMEM-EG-ST-CRT-0001	Structural Design Criteria – Buildings and Structures
MD-101-SMEM-EG-AR-CRT-0001	Architectural Design Criteria
MD-101-SMEM-EG-ST-SPC-0001	Steel Fabrication
MD-101-SMEM-EG-ST-SPC-0009	Protective Coating of Structural Steelwork
MD-101-SMEM-EG-ME-SPC-0008	Protective Coating for Concrete
MD-101-SMEM-EG-FG-SPC-0003	Fire Protection Systems

4. GENERAL DESIGN REQUIREMENTS

4.1 General

The design requirements provided in this document are generally applicable to Fire Retardant Coatings for Structures. In circumstances where a project specification exists, the requirements of the project specification shall have precedence over the requirements summarized in this document. Should the Engineer consider a general specification to be deficient in any way the concern must be raised with the Employer for resolution.

Should it be required to develop a project specification for equipment or materials, the general requirements nominated in this document must be incorporated as a minimum.

The Engineer must familiarize himself with the contents of Specifications applicable to the design and execute the design in consideration of the applicable Specifications.

4.2 Supplier Drawing and Documentation

The design of the Fire-Retardant Coatings on Structures shall be detailed in drawings to meet the requirements of the Employer's specifications. The drawings shall be prepared based on:

- A detailed design that meets the statutory requirements, and suits the building layouts and the system performance, including the type of coatings and description
- A copy of any design, calculations, appropriate drawings and supporting documents shall be submitted to the Employer for approval prior to construction and application of the Fire-Retardant Coatings on Structures.

4.3 Safety, Reliability, Availability and Maintainability

The design shall ensure that the fire-retardant coatings for structures meet the standards of safety, reliability, maintainability and availability set out in this specification. The objectives shall be to minimize maintenance aspects such as:

- Costs
- Complexity
- Personnel requirements
- Maintenance errors.

4.4 Site Data

All materials, components and installations selected shall be of suitable design, manufacture and construction that they may function satisfactorily over the full range of climatic and atmospheric conditions normally encountered in the project site where the plant is to be located. The site conditions shall be defined in the applicable Site Data Specification.

4.5 Design Life

The design life for plant equipment shall be 30 years minimum or as specified in the specific Project Basis of Design.

4.6 Provision for Future Expansion

Provision for future expansion shall be as specified in the specific Project Basic Engineering Design Data.

4.7 Standardization

To facilitate procurement, maintenance and spares holdings, design shall be standardized throughout the facility. Where practical, in all areas, the type of fire-retardant coatings of the same Manufacturer and type shall be used.

If it is not possible to standardize throughout the facility an alternative product may be used after prior approval from the Employer.

4.8 Quality Assurance

The fire-retardant coatings, where appropriate, shall be either UL listed and/or FM tested and approved for its intended use.

UL listed and/or FM tested and approved fire-retardant coating products is the industry benchmark and the minimum accreditation acceptable. The fire-retardant coatings shall be applied in accordance with the vendor recommendation such that compliance with the approved listing is not compromised.

The Fire Protection Systems and components shall be manufactured by an ISO 9001 certified company and meet the requirements of ISO 9001.

The Contractor shall comply with the Project Quality Requirements for specific QA/QC requirements.

5. REQUIREMENT OF FIRE-RETARDANT COATINGS FOR STRUCTURES

5.1 Legislative Requirements

The purpose of fire-retardant coatings for structures is to ensure an identified building or structure is designed and provided with a sufficient fire resistance level to ensure the fire-spread is limited to the index rating of 25 or less in accordance with NFPA 255, SAF-3 and SAF-9.

The Employer's preferred plant buildings material of construction shall be noncombustible materials, such as concrete, masonry and steel. All interior finishing materials and furniture shall be made of noncombustible material. Where this is not possible, alternative materials are allowed after approval from the Employer and shall have fire retardant coatings applied to the finishing materials to comply with NFPA 703.

The SAF-9 Section 4 specifies that the Engineer shall ensure that buildings and structures in a project plant are provided with fire resistance in accordance with API RP 2001. Provision of fire resistance rated construction shall prevent equipment and support structure failure during a fire.

Besides compliance to the SAF-9 Section 4, the Engineer shall ensure the use of construction components and material also comply to the fire-resistant rating as detailed in Section 4A and section 4B of SBC 801.

5.2 Fire Risk Assessment

The Engineer shall fulfill all the applicable statuary and legislative requirements in Section 5.1 as the minimum design standard. In addition, the Engineer is required to carry out fire risk assessment as detailed in this section (Section 5.2).

Where not required by the statuary and legislative requirements, the application of fire-retardant coatings on building and structure shall be subjected to a fire risk assessment. The Engineer shall identify and document any specific area or building or structure that is subject to high fire risk involving life & safety of operators.

The Engineer shall carry out risk assessment on the identified areas to determine the need for extra passive fire protection approach (e.g. provide higher fire resistance rating on building or structure than required by the SBC or SAF).

The risk assessment shall be prepared in accordance with the methodologies as described in the International Fire Engineering Guidelines. The Contractor shall ensure that all relevant stakeholders for the Employer's project are involved in the fire safety engineering process.

The risk assessment shall be carried out in accordance with ISO 31000 involving all relevant stakeholders to identify and analyze the fire and safety risks in an area, and subsequently agree on a solution to reduce the identified fire and safety risk. The hierarchy of controls shall include elimination, isolation or provision of engineering controls to manage the fire and safety risk.

In addition to the fire risk assessment, the Engineer shall consider the Employer insurer's specific requirements on the fire-retardant coatings for structure/buildings as detailed in Section 1)a)3.2.4.

5.3 System Design

The application of fire-retardant coatings on structure or building shall be based on analysis of potential fire scenarios. Typically, fire retardant coatings are required on the support structures of all vessels, piping and other major equipment located in areas where potential release of flammable or combustible fluids may be present which may result in substantial property damage.

The application of intumescent paints, approved spray-type cement concrete and plaster, mastics or other fire-retardant coatings shall be performed by personnel trained and certified by the material Manufacturer.

The fire-retardant coating system shall be a proprietary item and listed for its purpose. The application of concrete (e.g. dense concrete) without any test approvals is not considered as a proprietary fire-retardant coating.

All building interior walls and ceiling shall have finishes in accordance with the flame spread and smoke-developed index classification in accordance with Section 7.11 of SBC 201 and to comply with the allowable limit detailed in Table 6.6.3 of SBC 801.

Where the allowable index classification is exceeded, the Engineer shall apply approved fire-retardant coatings, paints or solutions to the protected surfaces. Such applications shall comply with NFPA 703, and the required fire-retardant properties shall be maintained or renewed in accordance with the manufacturer's instructions.

Fire retardant coating may be used as a substitute for an automatic fixed fire protection system for exposure protection purposes. Fire retardant coating shall not be used as the primary means of fire protection system in place of a fixed fire protection system.

6. FIRE RETARDANT COATINGS FOR STEEL STRUCTURES

Steel columns and beams that are supporting the critical equipment and/or in areas containing flammable and combustible materials, shall be protected by an approved fire-rated coating system when required by code.

The purpose of the fire rated system for the steel members is to prevent the collapse of the structures which will damage the equipment and may add more fuel (lubrication oil, cabling, etc) to the fire.

The coating system chosen must meet a list of requirements which include, but is not limited to:

- Delay damage to the structural steel elements in a fire keeping them functional for a period of time, established from codes, standards, insurance requirements or hazard studies
- Have good mechanical properties to ensure the system is not easily damaged from impact
- Be resistant to humidity, moisture and water contact
- Have good resistance to any chemicals, for cases where chemical exposure is expected.
- Have good adhesion to the steel members
- Not cause health problems during application
- Have a long durability.

The fire rated system can be either a combination of a spray on cement-based material, plasters, mineral fibers or intumescent system. There is no preferred type, however the chosen system must not degrade from moisture.

The fire rated system shall be applied as per the manufacturer's requirements and recommendations.

The steel structure members being protected shall be cleaned to the degree recommended by the Manufacturer of the passive fire system. As a minimum, oil, grease and obvious dirt should be removed with dry rags.

If a spray on fire rated cement is selected as the method of fire rating, the spray equipment used to apply the cement shall be approved for use by the fire rated coating manufacturer and shall be operated as per the spray equipment's operational instruction guide.

The necessary dry thickness of the coating must be achieved over the entire protected section as per the manufacturer's requirements. During times of high humidity in the air, ambient temperatures must be controlled, or a reinforced air movement (ventilator) may be utilized to help

to accelerate the drying of the coating. Intumescent coatings must not be installed when relative humidity is outside the limits established by the Manufacturer.

All steel elements that are critical for the stability of the structure under vertical loads shall be fire rated. For determining which members are critical for such stability, Engineer shall perform a stability analysis of the structure. Transient loads (e.g. wind, seismic) shall be excluded from such stability analysis. Based on the outcome of the stability analysis, the Engineer shall supply structural drawings and information to assist with identifying the critical elements.

The extent of the fire rating shall be as per the manufacturer's recommendations and shall be approved by the Employer.

The fire rated system shall be UL or FM Global approved and listed for use in the specific intended application.

The fire rated system shall be approved by the Employer prior to use.

Proprietary fireproofing products as mentioned in Table 6.1 may be used

Table 6.1- Fireproofing Coatings

S.NO.	Product	Description	Proprietor
1	Pyrocrete 241HY	High-Density Cementitious SFRM	Carboline Co., Fireproofing Products Div
2	Monokote Type Z-156	High-Density Cementitious SFRM	Grace, W. R. & Co. - Conn., Construction Products Div
3	Pyrocrete 239	Medium-Density Cementitious SFRM	Carboline Co., Fireproofing Products Div
4	Monokote Type Z146	Medium-Density Cementitious SFRM	Grace, W.R. & Co. - Conn., Construction Products Div
5	Basecoat, A/D Firefilm III and Colorcoat	Fire-Resistive, Water-Based, Intumescent Mastic Coating Material	Carboline Company, Fireproofing Products Div
6	Basecoat, Thermolag E100 and Colorcoat	Fire-Resistive, Epoxy-Based, Intumescent Mastic Coating Material (Shop Applied)	Carboline Company, Fireproofing Products Div
7	Basecoat, Thermolag E100 and Colorcoat	Fire-Resistive, Epoxy-Based, Intumescent Mastic Coating Material	Carboline Company, Fireproofing Products Div

The products used shall meet the requirements of the intended use as certified by the Manufacturer. Equivalent products shall be got approved by the Employer prior to use.

7. FIRE RETARDANT-TREATED WOOD

This section is applicable for the fire retardant-treated wood.

When used, all interior fire retardant-treated wood shall have a moisture content not exceeding 28% when tested in accordance with ASTM D 3201.

All wood structural panel and lumber shall be tested in accordance with requirements in Section 4.2 of NFPA 703.

Fire retardant–treated wood shall have a moisture content of 19% or less for lumber and 15% or less for wood structural panels before use.

Fire retardant-treated wood that is air-dried after treatment shall be protected from the weather. The Engineer shall only use fire retardant-treated wood identified as 'exterior grade' for installations that are exposed to wet or damp weather.

Fasteners, anchors and hardware that are hot dip galvanized or stainless steel shall be used to minimize corrosion potential.

8. FIRE RETARDANT COATINGS FOR BUILDING STRUCTURES

Some fire-retardant paints have intumescent properties causing the paint to form an insulating blanket when exposed to fire. This retards surface ignition and reduces the burning rate of the coated side of the combustible material.

Fire-retardant cement, gypsum, concrete or plaster that are applied on the building structure by the method of spraying shall be approved and listed based on the tests carried out in accordance with NFPA 255 and NFPA 705.

Fire-retardant coatings will maintain a noncombustible surface and effectively reduce the burning rate of the below combustible surface for a period of approximately 10 to 15 minutes.

When fire retardant-treated wood is not used, or the selected wood fails to achieve the flame spread limit set in the NFPA 703 and this specification, the wood shall have fire retardant coatings applied to it as per this section.

A fire-retardant coating shall not be used for unprotected outdoor installations unless labeled for such installations.

When applied on a structure, the fire-retardant coatings shall be protected against mechanical damage. Coatings that are exposed to high vehicle traffic or frequent equipment lifting areas shall be protected by installation of bollards or guarding.

Where a section or the entire area of the fire-retardant coatings undergoes structure modification work, the Engineer shall reapply the fire-retardant coating on the affected area. The classification of fire-retardant coatings shall be applied at the rates of coverage and to the applicable substance, building material, or species of wood indicated on the test report.

The fire-retardant coatings shall be applied in accordance with the manufacturer's directions supplied with the container.

The fire-retardant coatings shall be applied by licensed operator recommended by the paint/coating manufacturer. The operator who applied the fire-retardant coatings shall certify their work as being in conformance with the Manufacturer's directions for application.

A fire-retardant coating shall not be coated over with any material unless both the fire-retardant coating and the overcoat have been tested as a system and are found to meet the requirements of a fire-retardant coating.

Non-cementitious-based coatings shall not be used for high-humidity occupancies or for certain occupancies where combustible dust and other organic residues, such as oil, solvents, etc. may accumulate.

All fire-retardant coatings shall be tested either by NFPA 255, ASTM E 84 or UL 723, and listed for its intended use.

Where fire retardant coatings are to be subjected to sustained humidity of 80% or more or exposure to the weather, certification by a testing laboratory shall be required to indicate that there is no increase in listed classification is required as described in ASTM D 2898.

Fire retardant coatings shall possess the desired degree of permanency and shall be maintained to retain the effectiveness of the treatment under the service conditions encountered in actual use.